SENTIMENT ANALYSIS OF X DATA USING VADER

Introduction:

X is a widely used social media platform where users express their opinions on various topics. Sentiment analysis helps in understanding public opinion by classifying text into positive, negative, or neutral sentiments. This project analyzes sentiments in X posts using the VADER Sentiment Analyzer.

Objective:

The objective of this project is to analyze sentiments of posts on X using VADER and compare the predicted sentiments with the actual labeled dataset.

Dataset Description:

The dataset consists of two columns:

- **clean_text**: Preprocessed text of posts
- **category**: Sentiment label
  - 1 → Positive
  - 0 → Neutral
  - -1 → Negative

Methodology:

- The dataset was loaded using Pandas.
- VADER Sentiment Analyzer was applied to compute sentiment scores.
- A threshold of ±0.05 was used to classify sentiments.
- Predicted sentiments were compared with actual labels.
- Visualizations such as bar charts, pie charts, and confusion matrix were used for analysis.

Tools Used:

- Python
- Pandas
- NLTK (VADER Sentiment Analyzer)
- Matplotlib
- Seaborn

Results

Sentiment Distribution (Actual Data)

- Positive (1): 72,250 (**44.3%**)
- Neutral (0): 55,213 (**33.9%**)

- Negative (-1): 35,510 (**21.8%**)

Sentiment Distribution (VADER Prediction)

- Positive (1): 76,227 (**46.8%**)
- Neutral (0): 50,935 (**31.2%**)
- Negative (-1): 35,818 (**22.0%**)

Model Accuracy

The VADER model achieved an accuracy of **56.79%** in predicting sentiments compared to actual labels.

The results show that positive sentiment is dominant in both actual and predicted data. The VADER model slightly overestimates positive sentiment while maintaining similar distribution for negative sentiment.

Visualizations:

1. Bar Chart

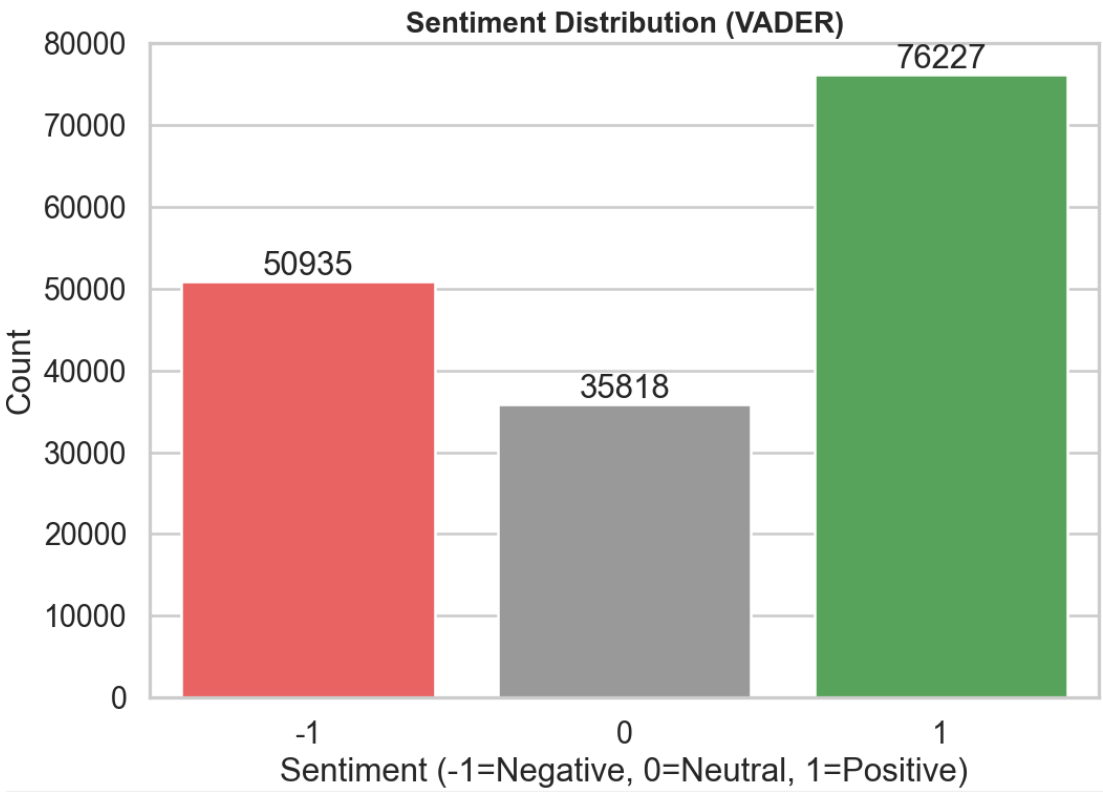

Figure 1: Bar chart showing distribution of predicted sentiments.

2. Pie Chart

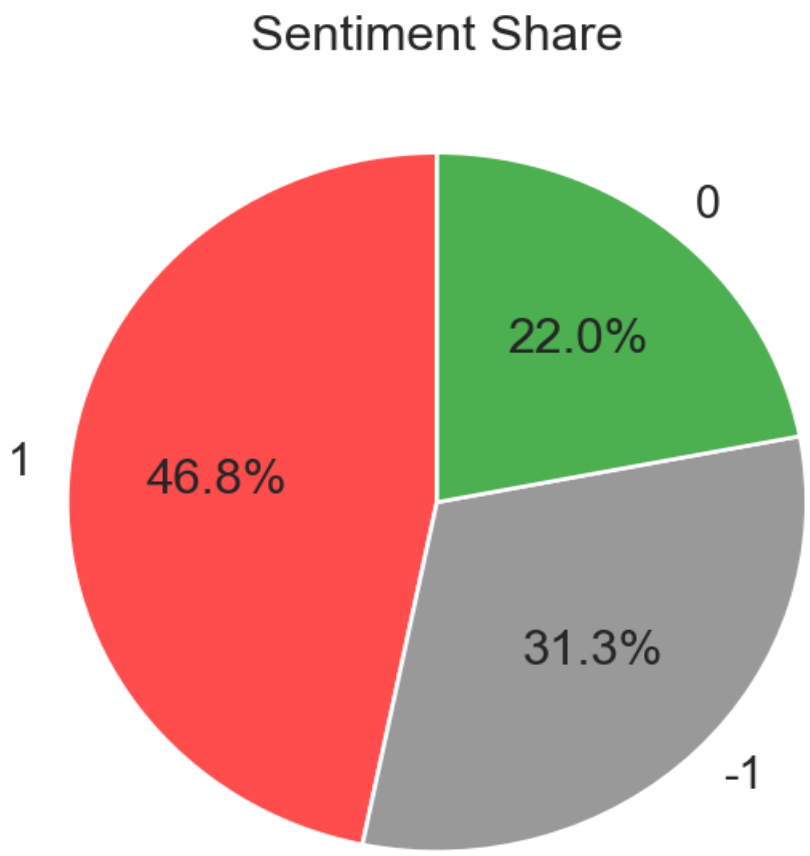

Figure 2: Pie chart showing proportion of sentiment categories

3. Confusion Matrix

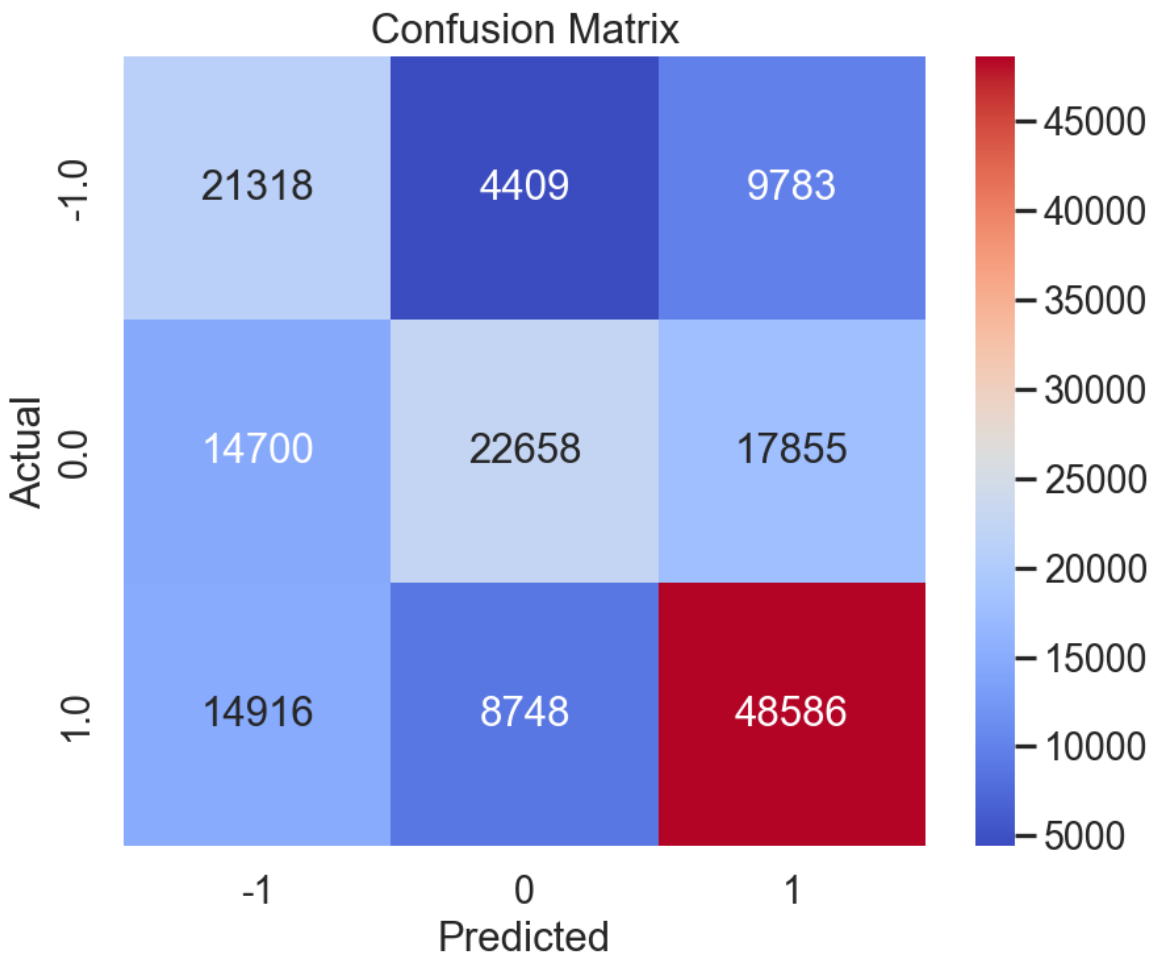

Figure 3: Confusion matrix comparing actual vs predicted sentiments

4. Score Distribution Graph

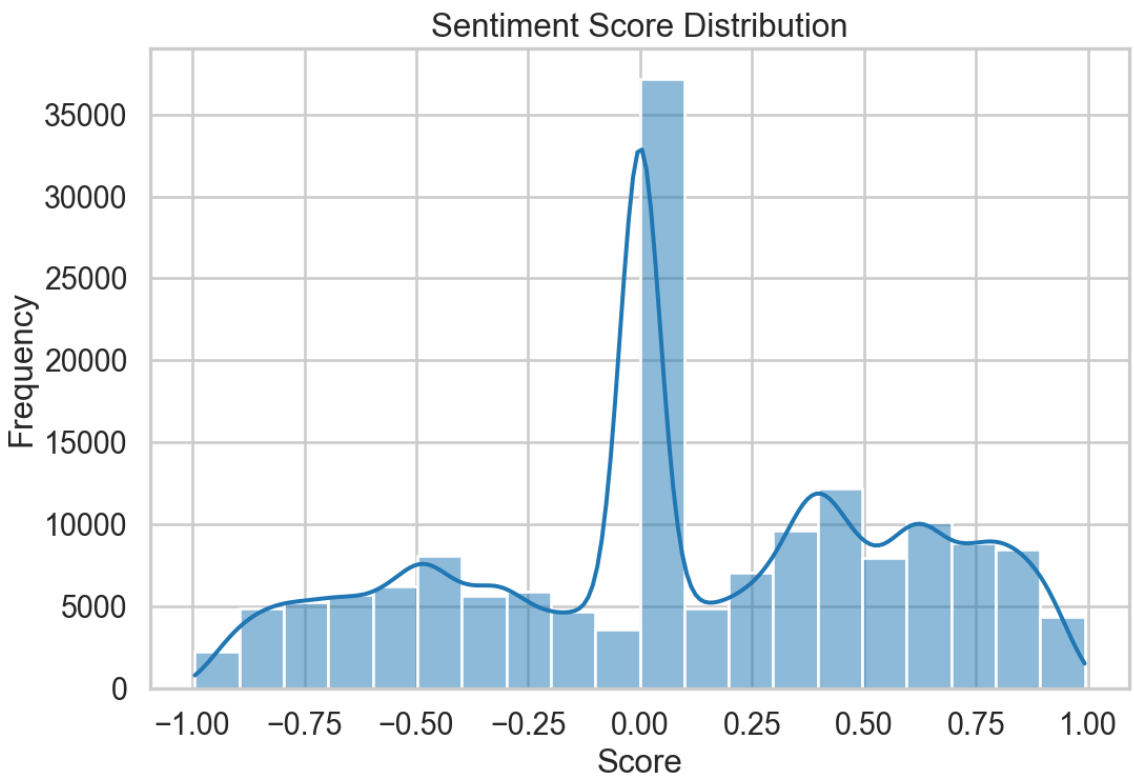

Figure 4: Distribution of sentiment scores across dataset.

The bar chart and pie chart show that positive sentiments dominate the dataset. The confusion matrix indicates that the model performs well in identifying positive sentiments, with minor misclassifications in neutral and negative categories.

Visualization Insights:

- The bar chart shows the distribution of predicted sentiments.
- The pie chart highlights the proportion of each sentiment category.
- The confusion matrix provides a comparison between actual and predicted values, showing how well the model performs across different sentiment classes.
- The sentiment score distribution graph shows how sentiment values are spread across the dataset.

Conclusion:

The analysis demonstrates that VADER is effective in identifying sentiments in social media text. The model provides a good approximation of actual sentiment labels and helps in understanding overall public opinion trends. This approach can be useful for monitoring discussions, identifying negative sentiment, and analyzing user behavior on X.

Applications:

- Social media monitoring
- Public opinion analysis
- Brand sentiment tracking
- Detection of negative or harmful content